

Learner Demographics and Course Enrollment Behavior Analysis on EduPro

A Data Analytics Research Paper

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An exploratory and descriptive analytics study of learner demographics, course preferences, enrollment behavior, segmentation, and decision-support insights using the EduPro online learning platform dataset.

Prepared in the structure and analytical spirit of the supplied reference research paper, adapted to the actual EduPro project evidence.

Abstract

This research paper presents a structured Data Analyst study of learner demographics and course enrollment behavior on the EduPro online learning platform. The study follows the organization of the supplied reference research paper while adapting its methodology to a descriptive, exploratory analytics project rather than inventing predictive-model results that are not present in the EduPro project. The analysis integrates four related tables—Users, Teachers, Courses, and Transactions—into a 10,000-row analytical dataset. The source tables contain 3,000 learner records, 60 teacher records, 60 course records, and 10,000 transaction records. The cleaned analytical dataset contains 25 columns and, at the documented quality-check stage, no missing values are reported. ■filecite■turn3file6■L375-L384■

The study examines learner age, gender participation, course category demand, course-level preferences, age-specific course choices, free-versus-paid enrollment, learner engagement frequency, spending, and course duration. The 26–35 group records the largest enrollment volume with 4,799 enrollments. Female learners account for 5,078 enrollments (50.78%) and male learners account for 4,922 (49.22%). Data Science is the leading course category with 916 enrollments, while Beginner courses lead the overall level distribution with 3,573 enrollments. Age-specific preferences are also visible: Artificial Intelligence leads below 18, Web Development leads among 18–25, and Data Science leads among 26–35. ■filecite■turn5file6■L385-L422■

Behavioral analysis shows an average of 3.33 courses per learner. Free courses receive more enrollments than paid courses in every age group. Average spending is highest for Beginner courses, while average course duration is relatively similar across levels. These results demonstrate how descriptive analytics can transform raw educational-platform records into interpretable evidence for learner segmentation, course planning, recommendation design, engagement strategy, and dashboard development. ■filecite■turn5file8■L458-L470■

Keywords: EduPro, Data Analytics, Learner Demographics, Enrollment Behavior, Exploratory Data Analysis, Educational Analytics, Course Preference, Learner Segmentation, Python, Pandas, Streamlit

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1. Introduction

Online learning platforms generate structured information about learners, courses, instructors, and transactions. When these sources are combined, data analysts can move beyond simple record keeping and examine who participates, what learners choose, how preferences differ across segments, and which parts of a course catalog receive the greatest demand. The supplied reference research paper similarly begins by establishing the business context, explaining the analytical challenge, and connecting data analysis to operational decision-making.

■filecite■turn4file0■L39-L60■

EduPro provides a suitable environment for this type of analysis because its project data are distributed across Users, Teachers, Courses, and Transactions tables. The project notebook describes objectives around user behavior, course engagement, revenue generation, and learning patterns. The transaction table is therefore treated as the central behavioral observation and is connected with demographic and course attributes.

■filecite■turn3file3■L223-L238■

The present study intentionally uses descriptive and exploratory analytics. Python-based tools including Pandas, NumPy, Seaborn, and Matplotlib are used for data inspection, cleaning, integration, aggregation, cross-tabulation, and visualization. This mirrors the reference paper's emphasis on understanding distributions and behavioral patterns before moving toward more advanced analytical methods.

■filecite■turn3file2■L152-L188■

The contribution of this project is not a predictive model; instead, it is an interpretable analytical framework that organizes the available EduPro evidence into demographic profiles, course-demand rankings, cross-segment comparisons, engagement indicators, and practical recommendations. The resulting insights are also suitable for presentation through the Streamlit dashboard developed for the project.

2. Problem Definition

The EduPro information is distributed across several related tables. Looking at each table separately can hide relationships between learner demographics, course attributes, and enrollment activity. The central analytical problem is therefore to construct an integrated view of the learner journey and determine whether enrollment behavior differs meaningfully across demographic and course dimensions.

■filecite■turn3file2■L156-L164■

The project addresses a practical decision-support problem: platform managers need to understand which learner groups are most active, which categories attract demand, which levels are preferred, whether free content is an important engagement entry point, and whether different demographic groups show distinct course-selection patterns.

Research gap: The supplied reference paper identifies a broader gap between raw data/model output and actionable business insight. For EduPro, the analogous gap is the absence of a single, interpretable view that connects demographic segmentation, course demand, level preference, and engagement behavior. The present project fills that descriptive-analytics gap by integrating the available tables and translating the resulting patterns into recommendations. This is an adaptation of the reference paper's framing, not a claim that the EduPro dataset contains a previously published research gap.

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3. Objectives and Research Questions

Objectives

- Build an integrated analytical dataset from Users, Teachers, Courses, and Transactions.
- Profile learners by age and gender.
- Measure enrollment concentration across age groups.
- Rank course categories using enrollment volume and a category popularity index.
- Measure overall course-level preference and compare level choices across age groups and gender.
- Identify the leading course category within each age segment.

- Compare free and paid enrollment behavior across age groups.
- Measure learner engagement using average courses per learner and active-user concentration.
- Compare average spending and course duration across course levels.
- Translate descriptive findings into practical course-planning, recommendation, engagement, and dashboard recommendations.

Research Questions

1. Which age group generates the most enrollments?
2. How is enrollment distributed by gender?
3. Which course categories and levels are most popular?
4. Do age groups show different category and level preferences?
5. How does free-versus-paid enrollment vary by age group?
6. How concentrated is enrollment activity among highly active learners?
7. Do average spending and course duration differ across course levels?

4. Literature Review

The literature review follows the conceptual role of the supplied reference paper: establish why the analytical method is appropriate, identify the value of interpretable analysis, and connect segmentation to practical decisions. The reference paper discusses a progression from interpretable statistical approaches toward more complex machine-learning approaches, while emphasizing that exploratory analysis is necessary to understand the structure of the data before advanced modeling. ■filecite■turn4file0■L98-L127■

4.1 Educational and Behavioral Analytics

Educational analytics uses learner and course data to identify participation patterns, preferences, and engagement signals. In an online-learning context, demographic and behavioral variables can be combined to understand demand across learner segments. For EduPro, enrollment is used as the principal observable measure of participation, allowing the analysis to remain grounded in the project data rather than introducing unsupported learning-outcome measures.

4.2 Exploratory Data Analysis

The supplied reference paper explicitly describes EDA as a preliminary stage for revealing data structure, distributions, and behavioral patterns before more advanced modeling. EduPro adopts this principle through univariate analysis of age, gender, category, level, and course type, followed by bivariate and cross-tabulation analysis such as age $\times$ level, gender $\times$ level, and age $\times$ category. ■filecite■turn4file0■L132-L148■

4.3 Segmentation and Personalization

Segmentation is another major theme in the reference paper. Its analysis uses behavioral segmentation to distinguish customer personas and translate differences into targeted strategies. EduPro adapts the same analytical logic using age, gender, course category, course level, and course type as learner-segmentation dimensions. Because the EduPro project does not contain a validated clustering model, the paper uses descriptive segments rather than claiming formal machine-learning personas. ■filecite■turn3file2■L182-L188■

4.4 Research Gap

The project-level gap is practical rather than predictive: without integration and segmentation, enrollment counts alone cannot explain how demand varies between learner groups. This study contributes an interpretable framework combining demographic profiling, course-demand ranking, cross-tabulations, engagement indicators, and decision recommendations. It also identifies a path for future predictive work without presenting unperformed

models as results.

5. Methodology

The methodology is adapted from the reference paper's sequence of data acquisition, data description, cleaning, EDA, feature engineering, evaluation, and results interpretation. Because EduPro is a descriptive analytics project, predictive-model development, model tuning, SHAP analysis, and classification performance testing are not claimed. Instead, the corresponding methodological role is fulfilled by aggregation, ranking, segmentation, and descriptive evaluation.

5.1 Data Acquisition

The project notebook works with four related source tables: Users, Teachers, Courses, and Transactions. Users provides learner information; Courses provides course characteristics; Teachers provides instructor attributes; and Transactions provides enrollment/transaction observations. The documented source dimensions are 3,000 Users rows, 60 Teachers rows, 60 Courses rows, and 10,000 Transactions rows. [filecite turn3file3 L223-L235](#)

5.2 Data Description

Source Table	Rows	Columns	Analytical Role
Users	3,000	5	Learner demographic information
Teachers	60	7	Instructor and teaching attributes
Courses	60	8	Category, type, level, price, duration, rating
Transactions	10,000	7	Enrollment and transaction behavior
Cleaned analytical dataset	10,600	25	Integrated analysis and dashboard source

The central analytical dataset is transaction-level. Each transaction is enriched by matching learner, course, and teacher attributes, creating a unified record for demographic and course-preference analysis.

[filecite turn3file3 L223-L238](#)

5.3 Data Cleaning and Pre-processing

The project workflow includes data inspection, datatype checking, missing-value checking, standardization of learner fields, creation of an age-group variable, and export of a cleaned analytical dataset. The documented quality check reports zero missing values in the integrated fields at that stage. The integrated dataset contains 24 columns before learner-field standardization and the exported cleaned dataset contains 25 columns.

[filecite turn3file6 L375-L384](#)

The preparation stage is important because descriptive statistics and visualizations are only meaningful when identifiers, categorical fields, and numerical measures are consistently represented. Age is transformed into analytical bands—Below 18, 18–25, and 26–35—to support segment-level comparisons.

5.4 Data Integration

Transactions is merged with Users on UserID, Courses on CourseID, and Teachers on TeacherID. This creates a combined analytical view linking each enrollment observation with learner demographics, course characteristics, and teacher information. The cleaned output is then used for downstream analysis and Streamlit dashboard development. [filecite turn3file6 L375-L384](#)

5.5 Exploratory Data Analysis

- Univariate analysis: age distribution, gender participation, category counts, course-level counts, and course-type counts.
- Bivariate analysis: age group × category, age group × level, gender × level, and age group × course type.

- Ranking analysis: course categories ordered from highest to lowest enrollment.
- Distribution analysis: comparison of free and paid enrollment volumes.
- Behavioral analysis: average courses per learner, active-user concentration, average spending, and course duration.
- Visualization: bar charts, ranked horizontal bars, heatmaps, and grouped comparisons.

5.6 Feature Engineering

Feature engineering in this descriptive project is analytical rather than predictive. The principal derived features are age groups and the Category Popularity Index. The popularity index is defined as the category's enrollment count divided by total enrollment and multiplied by 100. Learner engagement is summarized as average courses per learner. These derived measures make raw transaction counts easier to compare and interpret.

Metric	Definition	Purpose
Total Enrollments	Count of transaction/enrollment records	Overall engagement volume
Age Group Enrollment	Count by analytical age band	Demographic reach
Gender Participation %	Gender enrollment / total enrollment $\times$ 100	Participation balance
Category Popularity Index	Category enrollments / total enrollments $\times$ 100	Relative course demand
Level Preference %	Level enrollments / total enrollments $\times$ 100	Skill-level demand
Average Courses per Learner	Total enrollments / unique learners	Repeat engagement
Active-Learner Concentration	Share of enrollments contributed by top active learners	Concentration of activity
Average Spending	Mean transaction amount by level	Commercial comparison
Average Duration	Mean course duration by level	Course-design comparison

5.7 Analytical Evaluation Metrics

5.8 Ethical and Practical Considerations

The analysis is intended for academic and decision-support use. Personal contact fields present in source structures are not used in the reported visualizations or recommendations. Any future production deployment should apply appropriate access control, data minimization, and privacy safeguards. The paper also distinguishes descriptive association from causation and does not infer that a demographic group causes a particular course choice.

6. Results and Findings

This chapter mirrors the reference paper's results-first structure by beginning with data quality and distribution analysis and then moving through increasingly detailed segment comparisons. The figures and tables are reconstructed from the recorded EduPro project outputs.

6.1 Data Quality and Distribution Analysis

The integrated dataset contains 10,000 analytical records. Learner age ranges from 15 to 35 years, with a mean of approximately 24.97 and a median of 25. The three project age bands are Below 18, 18–25, and 26–35.

filecite turn3file6 L385-L390

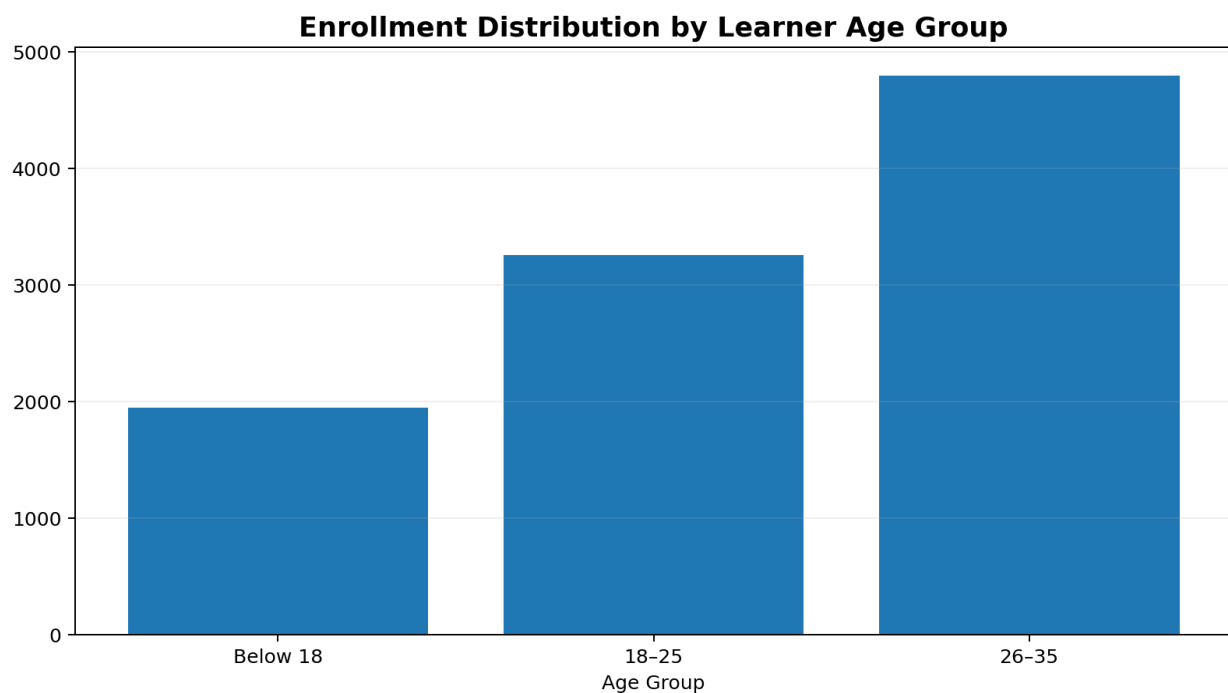


Figure 1. Enrollment Distribution by Learner Age Group

The 26–35 segment is the largest enrollment group, with 4,799 enrollments. The 18–25 group contributes 3,255, while learners below 18 contribute 1,946.

6.2 Learner Demographic Profile

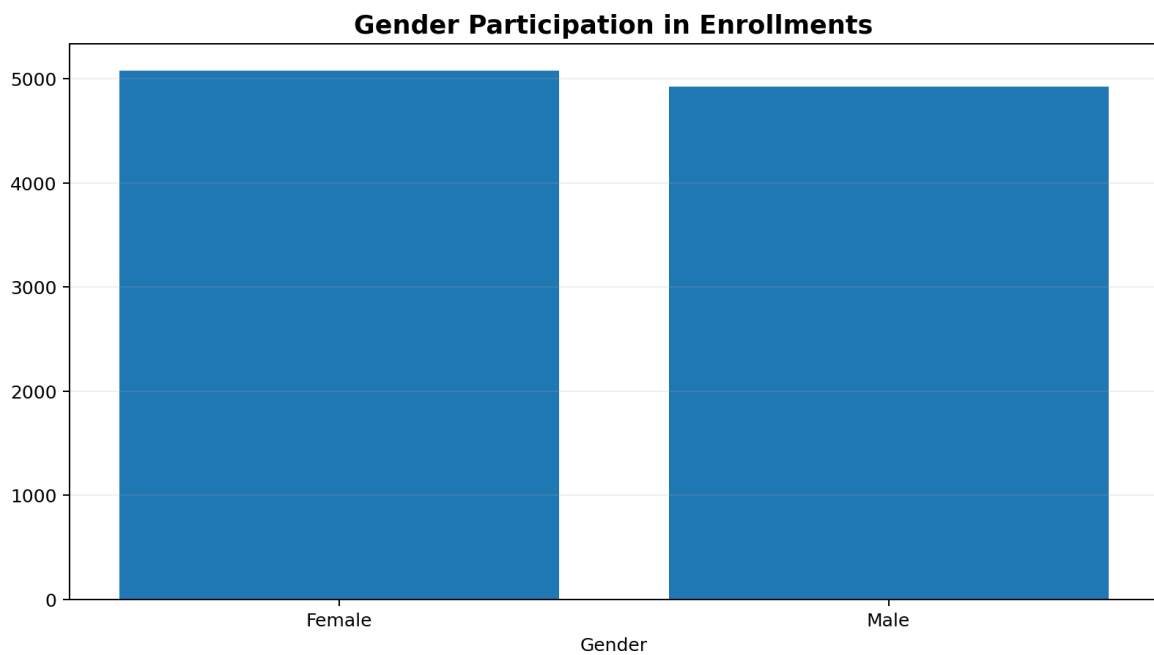


Figure 2. Gender Participation in Enrollments

Female learners account for 5,078 enrollments (50.78%) and male learners account for 4,922 (49.22%). The difference is 1.56 percentage points, indicating near-balanced participation.

Gender	Enrollments	Share
Female	5,078	50.78%
Male	4,922	49.22%
Total	10,000	100.00%

6.3 Course Category Popularity

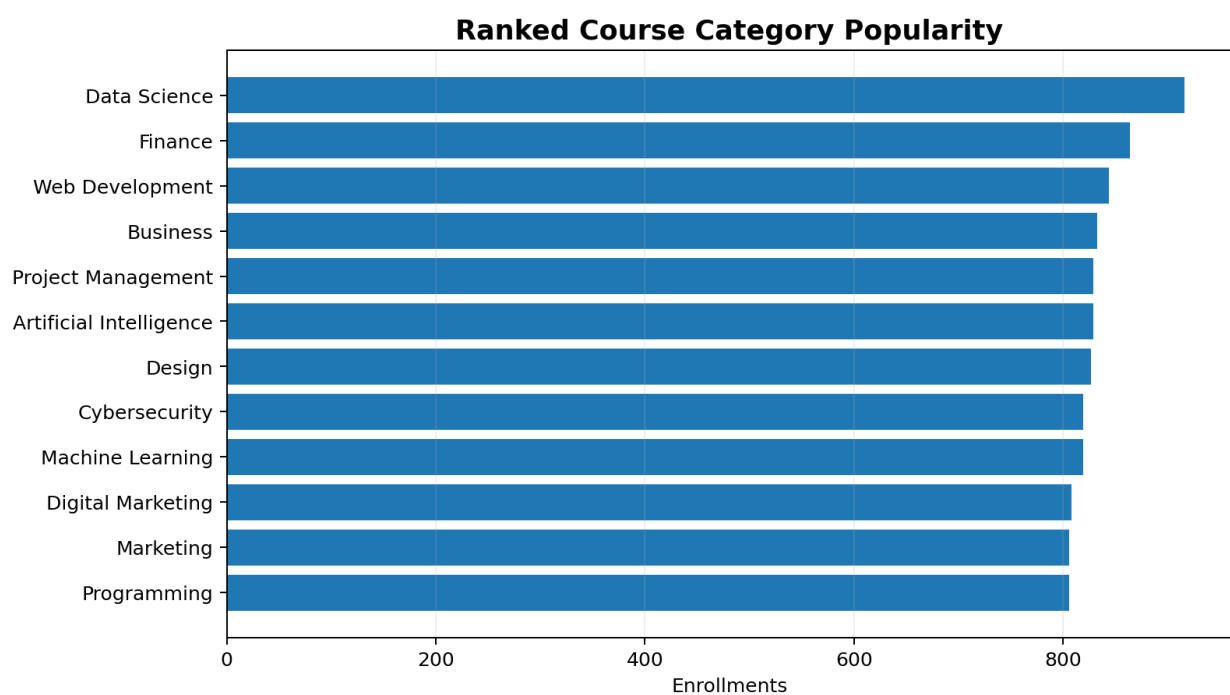


Figure 3. Ranked Course Category Popularity

Data Science is the most popular category with 916 enrollments and a popularity index of 9.16%. Finance follows with 864 and Web Development with 844. Marketing and Programming are the lowest-ranked categories at 806

each. The relatively narrow range indicates that demand is distributed across the catalog rather than concentrated in one category. ■filecite■turn5file6■L393-L412■

Rank	Category	Enrollments	Popularity Index
1	Data Science	916	9.16%
2	Finance	864	8.64%
3	Web Development	844	8.44%
4	Business	833	8.33%
5	Project Management	829	8.29%
6	Artificial Intelligence	829	8.29%
7	Design	827	8.27%
8	Cybersecurity	819	8.19%
9	Machine Learning	819	8.19%
10	Digital Marketing	808	8.08%
11	Marketing	806	8.06%
12	Programming	806	8.06%

6.4 Course Level Preference

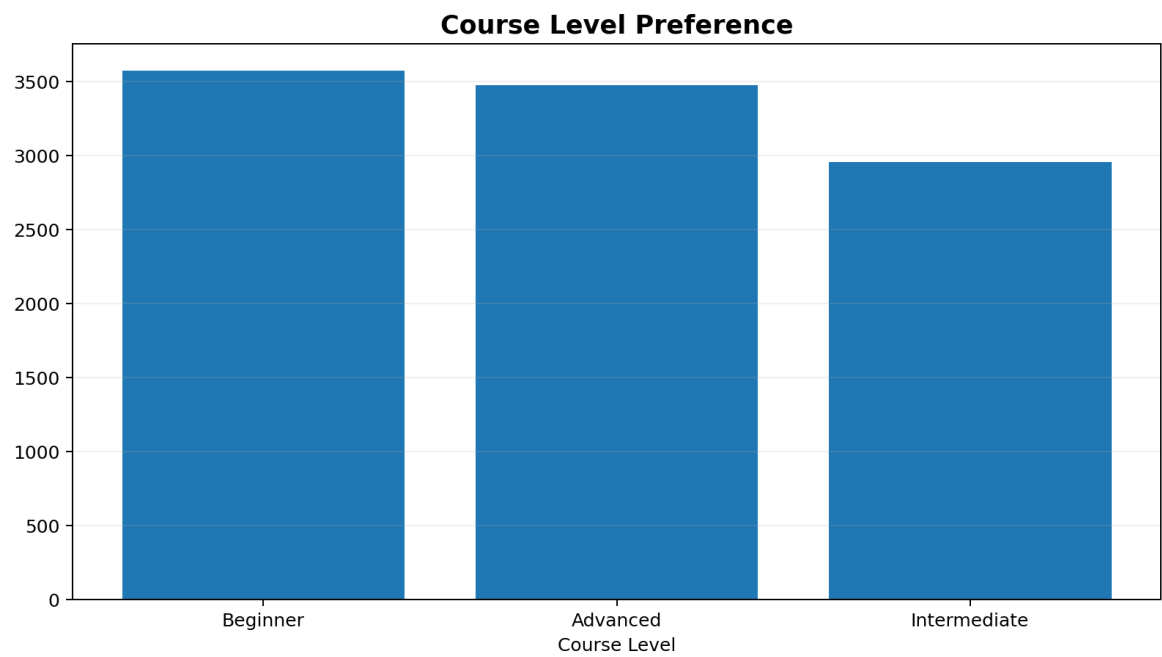


Figure 4. Course Level Preference

Beginner courses have the highest overall enrollment at 3,573 (35.73%), followed closely by Advanced at 3,475 (34.75%). Intermediate courses account for 2,952 (29.52%). The small difference between Beginner and Advanced indicates demand for both entry-level and advanced learning.

Level	Enrollments	Preference
Beginner	3,573	35.73%
Advanced	3,475	34.75%
Intermediate	2,952	29.52%

6.5 Course Preference by Age Group

Age-specific category preferences are distinct. Artificial Intelligence leads among learners below 18 with 176 enrollments; Web Development leads among learners aged 18–25 with 300; Data Science leads among learners aged 26–35 with 459.

Age Group	Leading Category	Enrollments
Below 18	Artificial Intelligence	176
18–25	Web Development	300
26–35	Data Science	459

6.6 Course Level Preference by Age Group

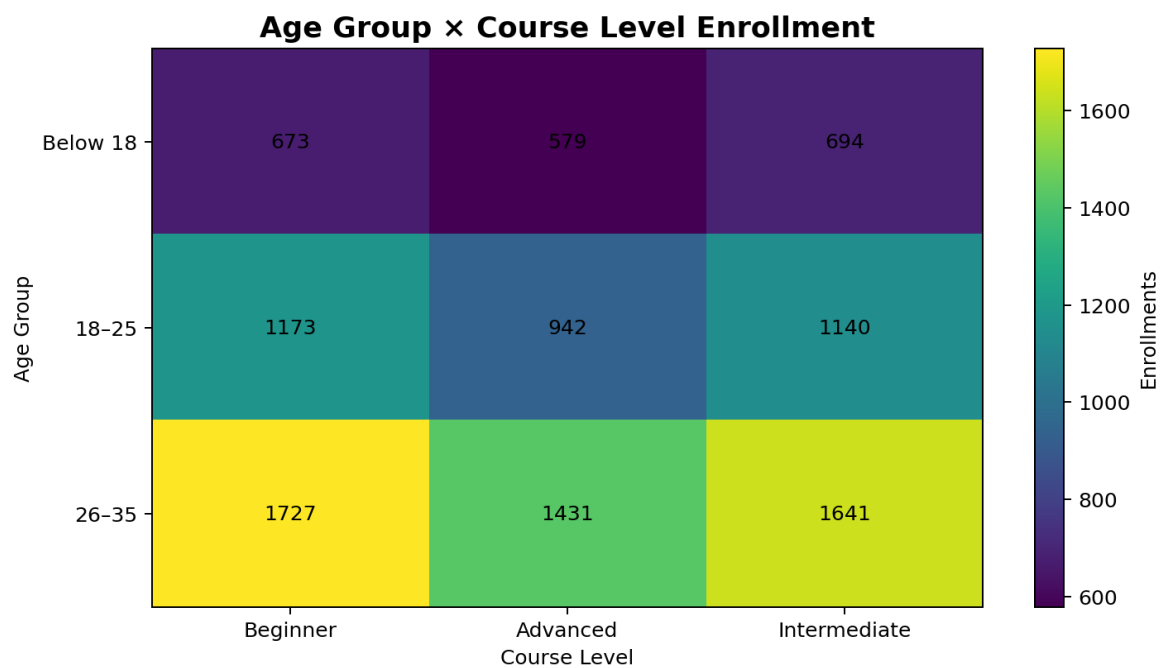


Figure 5. Age Group × Course Level Enrollment

Age Group	Beginner	Intermediate	Advanced
Below 18	673	579	694
18–25	1,173	942	1,140
26–35	1,727	1,431	1,641

Learners below 18 show Advanced as their most-enrolled level (694). The 18–25 group prefers Beginner (1,173), and the 26–35 group also prefers Beginner (1,727). This demonstrates that age does not map to course difficulty in a simple linear progression. ■filecite■turn5file8■L450-L456■

6.7 Gender × Course Level

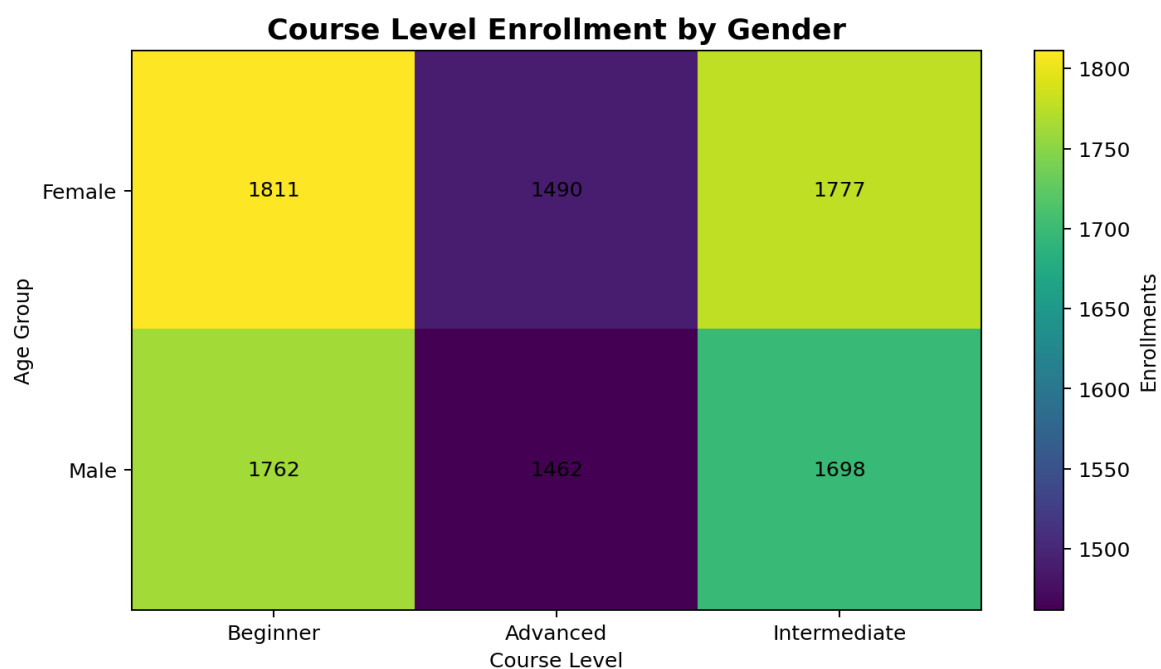


Figure 6. Course Level Enrollment by Gender

Gender	Beginner	Intermediate	Advanced
Female	1,811	1,490	1,777
Male	1,762	1,462	1,698

Beginner is the largest level for both genders, and the overall distributions are similar. This supports using gender as a monitoring and descriptive dimension rather than treating it as the primary recommendation variable.

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6.8 Free versus Paid Enrollment

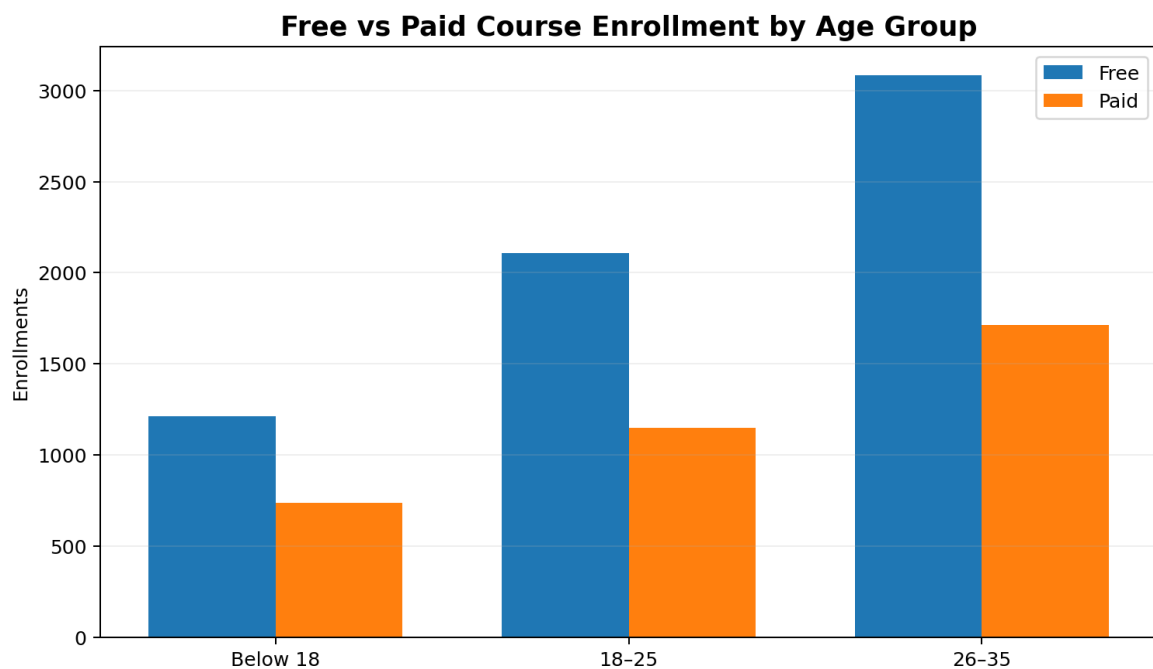


Figure 7. Free vs Paid Course Enrollment by Age Group

Age Group	Free	Paid	Free Share
Below 18	1,210	736	62.07%
18–25	2,108	1,147	64.75%
26–35	3,085	1,714	64.30%

Free courses lead paid courses in every age group. The difference is particularly visible in the two older segments, where free enrollment is roughly two-thirds of age-group enrollment. ■filecite■turn5file8■L458-L461■

7. Behavioral and Learner Segmentation Analysis

The reference paper moves from descriptive findings toward customer segmentation and campaign-fatigue interpretation. EduPro does not contain the target variables or modeling pipeline required to reproduce that banking analysis. Instead, this chapter adapts the same progression using observable learner engagement, concentration, spending, duration, and descriptive segment profiles.

7.1 Engagement Frequency

The project reports an average of 3.33 courses taken per learner. This indicates that learner participation extends beyond one-course activity and creates an opportunity for repeat-learning and pathway analysis.

■filecite■turn5file8■L462-L464■

7.2 Active-Learner Concentration

The ten most active users each have 16 enrollments. Together, these ten users account for 1.6% of the 10,000 enrollments. This indicates that enrollment activity is broadly distributed rather than being dominated by a very small group of learners. ■filecite■turn5file8■L463-L466■

Indicator	Result	Interpretation
Average courses per learner	3.33	Repeat engagement exists
Top active-user enrollments	16 each	High-engagement learners are identifiable
Top 10 share	1.6%	Overall enrollment is broadly distributed

7.3 Spending and Duration

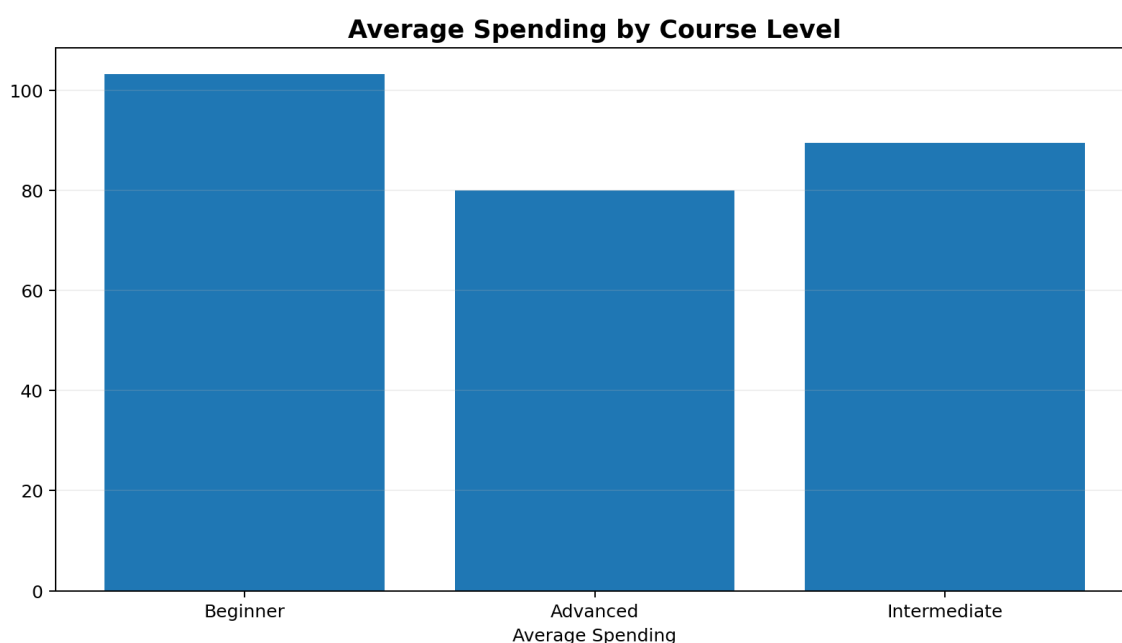


Figure 8. Average Spending by Course Level

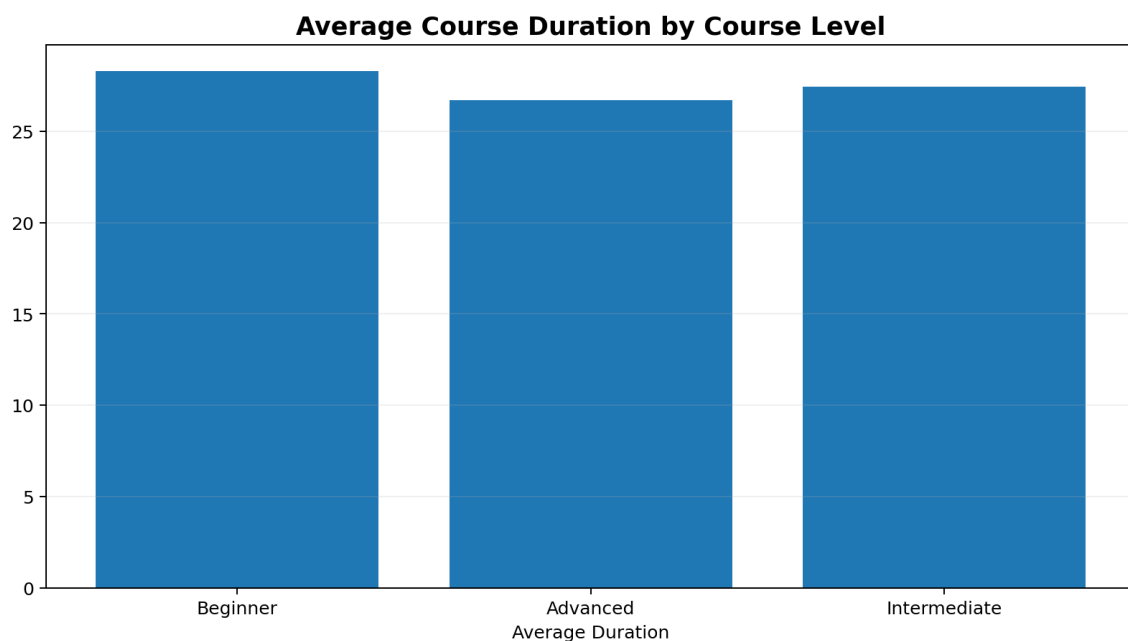


Figure 9. Average Course Duration by Course Level

Level	Average Spending	Average Duration
Beginner	103.28	28.31
Advanced	80.01	26.73
Intermediate	89.52	27.46

Average spending is highest for Beginner courses, while Advanced has the lowest average spending. Average duration is comparatively close across levels. Therefore, enrollment volume and average transaction value should be monitored as separate KPIs rather than treated as equivalent measures of performance.

filecite turn5file8 L467-L470

7.4 Learner Segment Interpretation

Descriptive Segment	Observed Pattern	Potential Analytical Use
Below 18	AI leads; Advanced level is highest	Youth-oriented discovery and advanced-interest analysis
18–25	Web Development leads; Beginner highest	Early-career skill discovery and pathway design
26–35	Data Science leads; Beginner highest	Professional upskilling and specialization analysis
Female	5,078 enrollments; Beginner highest	Participation monitoring
Male	4,922 enrollments; Beginner highest	Participation monitoring
Highly active learners	Top users reach 16 enrollments	Repeat-learning and retention analysis

These are descriptive personas or segments, not formal k-means clusters. This distinction is important because the reference paper uses an actual clustering method to create customer personas, whereas the EduPro evidence supports only rule-based descriptive segmentation. filecite turn4file0 L382-L390

8. Discussion

The discussion follows the reference paper's principle of interpreting empirical results and connecting them to operational implications. The emphasis is on what the EduPro patterns mean for course planning, learner experience, and analytics-driven decision support. ■filecite■turn4file0■L417-L429■

8.1 Interpretation of Demographic Patterns

The 26–35 segment is the largest contributor to enrollment volume, indicating strong participation from adult learners in the available data. However, age alone does not determine learning difficulty: the below-18 segment shows the highest Advanced enrollment within its group, while the two older groups prefer Beginner. This means recommendation systems should avoid assuming that older learners always seek advanced courses or that younger learners always begin at entry level.

8.2 Interpretation of Course Demand

Data Science leads the overall category ranking, but age-specific preferences reveal additional diversity. Artificial Intelligence is the leading category below 18, Web Development leads 18–25, and Data Science leads 26–35. Therefore, an overall popularity ranking is useful for catalog-level decisions, but it should not replace segment-level analysis. ■filecite■turn5file6■L423-L427■

8.3 Engagement and Commercial Implications

Beginner courses have both the highest enrollment and the highest average spending in the recorded outputs. At the same time, Advanced courses have nearly comparable enrollment volume but lower average spending. This demonstrates the value of combining engagement KPIs with commercial measures. Free courses dominate enrollment across all age groups, suggesting that free content is an important engagement entry point; however, the data do not establish whether free enrollment causes later paid conversion. ■filecite■turn5file8■L472-L485■

8.4 Implications for Dashboard Decision Support

The analysis naturally maps to an interactive dashboard: demographic overview, age-wise enrollment, gender-based preference analysis, course-category popularity, and filters for age group, gender, category, and level. The cleaned 25-column dataset can support such interactive KPI views. ■filecite■turn5file4■L257-L266■

9. Recommendations

- 1. Age-aware recommendations:** highlight Artificial Intelligence for below-18 learners, Web Development for 18–25, and Data Science for 26–35 while retaining broader choices.
- 2. Strengthen high-demand categories:** Data Science and Finance can receive additional depth, projects, and learning pathways because they lead overall category demand.
- 3. Use free courses as discovery entry points:** free content can support exploration and engagement before learners move toward paid specialization. This should be tested rather than assumed to cause conversion.
- 4. Avoid over-segmentation by gender:** participation is close to balanced and course-level distributions are similar; recommendation logic should prioritize learning intent and behavior.
- 5. Create level-progression pathways:** connect Beginner, Intermediate, and Advanced courses into coherent learning journeys.
- 6. Track engagement and commercial KPIs together:** combine enrollments, repeat courses, average spending, ratings, completion, and paid conversion in future dashboards.
- 7. Prioritize interactive dashboard exploration:** expose age, gender, category, level, and course-type filters so users can validate patterns for selected segments.
- 8. Extend the project toward predictive analytics:** once longitudinal and outcome variables become available, investigate recommendation, repeat-enrollment, paid-conversion, and learner-retention models.

These recommendations follow the reference paper's broader principle that analytical findings should be translated into operational decisions while avoiding unsupported causal claims. ■filecite■turn5file3■L216-L232■

10. Limitations

- The study is descriptive and identifies patterns and associations rather than causal relationships.
- Completion, watch time, assessment performance, learner satisfaction, and long-term retention are not included in the documented analysis.
- The three age bands are analytical choices; alternative boundaries could produce different segment results.
- The available project evidence represents a finite observation period, so long-term trends and seasonality cannot be established.
- Free-course dominance cannot be interpreted as proof of later paid conversion because longitudinal conversion data are unavailable.
- Descriptive learner segments should not be treated as formal machine-learning clusters.
- The study does not reproduce the predictive-modeling, model-performance, SHAP, or causal-analysis sections of the reference banking thesis because equivalent EduPro evidence was not provided.

These limitations preserve the reproducibility principle emphasized in the earlier version: the paper reports what the EduPro project supports and does not invent unavailable metrics or model results.

■filecite■turn5file3■L233-L239■

11. Conclusion

This research paper presented a complete descriptive Data Analyst study of learner demographics and course enrollment behavior on EduPro. Following the structure of the supplied reference research paper, the study moves from introduction and problem definition through literature review, methodology, data preparation, EDA, detailed results, behavioral segmentation, discussion, recommendations, limitations, and future work.

The analysis integrates 3,000 learner records, 60 course records, 60 teacher records, and 10,000 transaction records into a 10,000-row analytical dataset. The strongest findings are that the 26–35 group is the largest enrollment segment with 4,799 enrollments; female participation is 50.78%; Data Science is the leading category with 916 enrollments; Beginner is the leading overall level with 3,573 enrollments; and the average learner takes 3.33 courses. ■filecite■turn3file5■L343-L354■

The study also demonstrates the importance of segmentation: Artificial Intelligence leads below 18, Web Development leads at 18–25, and Data Science leads at 26–35. Free courses lead paid courses in every age group, while average spending and duration show that engagement and commercial value should be monitored together. These findings provide a practical foundation for learner recommendation, course planning, engagement strategy, and interactive dashboard design.

Overall, the project demonstrates how a Data Analyst can convert multi-table educational-platform data into a structured evidence base for decision-making. The next stage is not to assume that the descriptive relationships are causal, but to extend the dataset and test recommendations through longitudinal analysis, experimentation, and appropriate predictive methods.

12. Future Work

- Learner clustering using K-means or hierarchical methods after defining validated behavioral features.
- Course recommendation using learner-course interaction history.
- Repeat-enrollment prediction using longitudinal transaction sequences.
- Paid-conversion analysis to test whether free-course participation predicts later paid enrollment.

- Retention and completion analysis using watch time, completion, assessment, or activity logs.
- Time-based cohort analysis to identify changes in course demand and engagement.
- Explainable predictive models to connect recommendations with interpretable drivers.
- A/B testing of age-aware recommendations and free-to-paid learning pathways.

These extensions align with the earlier project's identified future directions—course recommendation, repeat-enrollment prediction, paid-conversion modeling, learner clustering, and longitudinal engagement analysis—while keeping them clearly separated from the current descriptive results.

■filecite■turn5file4■L274-L285■

13. References

The reference list below is based on the sources included in the supplied reference research paper and is used here for methodological framing rather than as evidence for EduPro-specific numerical results.

■filecite■turn4file0■L683-L827■

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Appendix A. Analytical Tables

A1. KPI Summary

KPI	Result	Interpretation
Total enrollments	10,000	Overall platform engagement volume
Largest age segment	26–35: 4,799	Largest demographic enrollment contribution
Female participation	50.78%	Near-balanced gender participation
Top category	Data Science: 916	Highest course-category demand
Top level	Beginner: 3,573	Highest overall level demand
Average courses/learner	3.33	Evidence of repeat-course participation
Top-10 active-user share	1.6%	Enrollment activity broadly distributed

A2. Age Group × Course Level

Age Group	Beginner	Intermediate	Advanced
Below 18	673	579	694
18–25	1,173	942	1,140
26–35	1,727	1,431	1,641

A3. Gender × Course Level

Gender	Beginner	Intermediate	Advanced
Female	1,811	1,490	1,777
Male	1,762	1,462	1,698

A4. Free vs Paid by Age

Age Group	Free	Paid
Below 18	1,210	736
18–25	2,108	1,147
26–35	3,085	1,714

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Appendix C. Project Traceability

Reference-paper component	EduPro paper implementation
Cover / Abstract / Keywords	Project title, author, Data Analyst course, study summary
Introduction	EduPro context and analytical motivation
Problem Definition	Integrated learner-course enrollment problem
Literature Review	EDA, segmentation, interpretability, research gap
Methodology	Data acquisition, description, cleaning, integration, EDA, feature engineering, metrics
EDA / Results	Demographics, category, level, age, gender, free/paid, spending, duration
Segmentation	Descriptive learner segments; no unsupported clustering claim
Discussion	Interpretation and operational implications
Recommendations	Age-aware recommendations, catalog planning, free-content strategy, dashboard KPIs
Limitations	Causal, temporal, outcome-data, and modeling limitations
Conclusion	Summary of findings and analytical contribution
Future Work	Recommendation, conversion, clustering, retention, longitudinal analysis
References	Sources carried from supplied reference paper
Appendices	Analytical tables and figure index

Important source note: the supplied reference paper contains extensive predictive-modeling, SHAP, ROC/AUC, clustering, campaign-fatigue, and model-performance sections. Those elements have not been falsely reproduced as EduPro results. Where the structure was relevant, it has been adapted to descriptive analytics using the actual EduPro project outputs. This preserves the reference paper's organization while keeping the paper evidence-based.